Experiment -2

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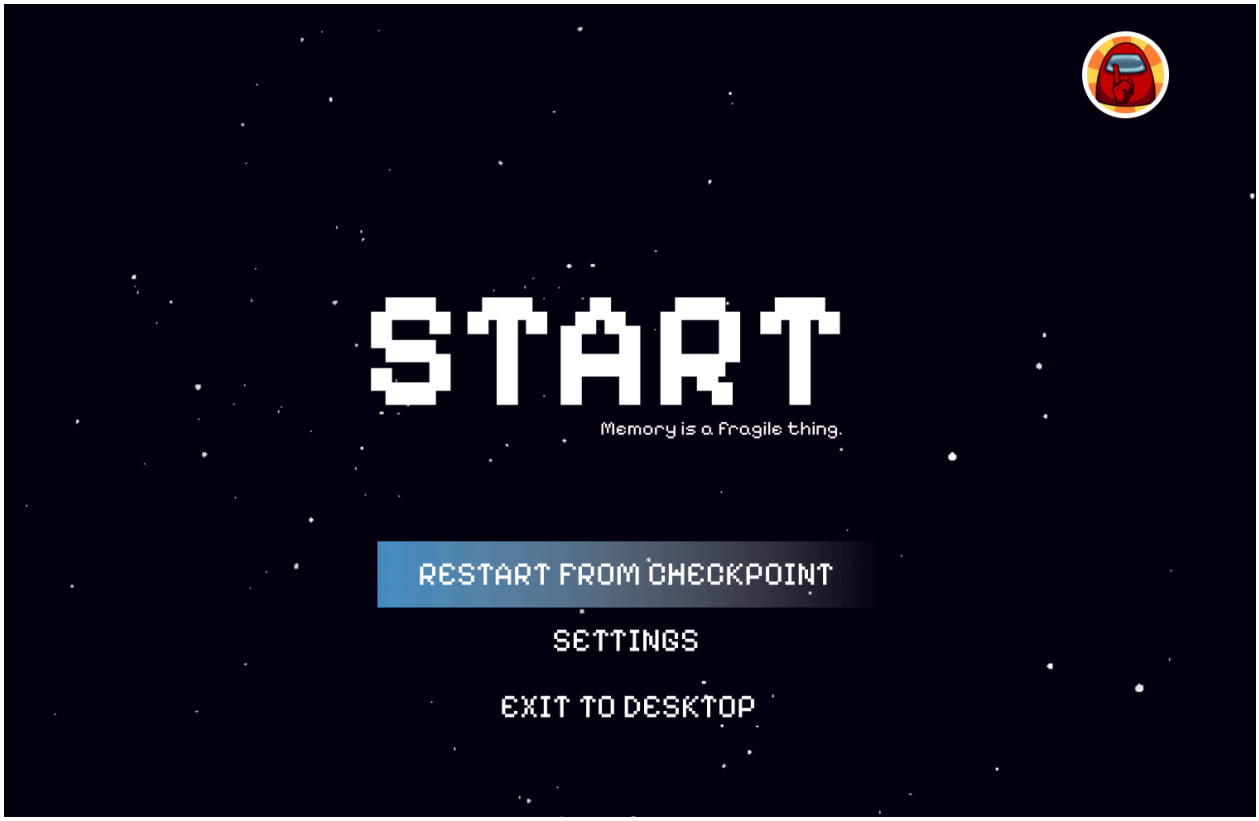
Design a UI where users recall visual elements (e.g., icons or text chunks). Evaluate the effect of chunking on user memory.

FRAME 1: INTRODUCTION PAGE (HOME SCREEN)



Analysis of the Introduction Screen

This screen serves as the entry point of the visual memory recall experiment and introduces the user to the game-based task in a simple and engaging manner.

1. Purpose of the Screen

- Introduces the user to the memory-based game environment.
- Sets the context for a visual recall task using icons inspired by a familiar game theme.
- Encourages the user to begin the experiment through a clear **START** option.

2. Chunking Strategy Applied

- Options such as **Start**, **Restart from Checkpoint**, **Settings**, and **Exit** are grouped vertically.
- Each option represents a distinct functional chunk, reducing decision complexity.
- Minimal text ensures users are not cognitively overloaded at the beginning.

3. Visual & UX Design

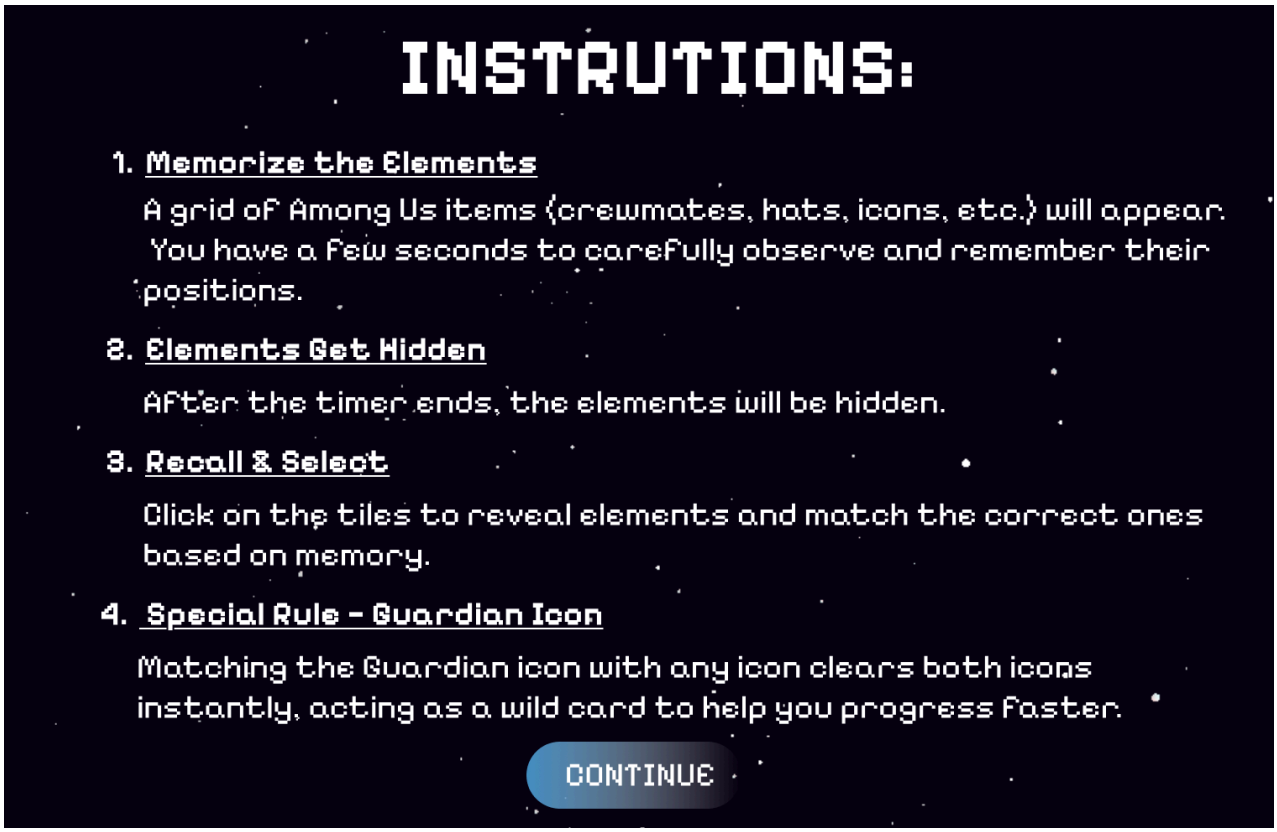
- Dark space-themed background minimizes distraction.
- Large pixel-style **START** text acts as a strong visual anchor.

- High contrast between text and background improves readability.
- Navigation elements are centrally aligned for easy access.

4. Cognitive Benefit

- Simple and chunked navigation reduces initial cognitive load.
- Users can focus on understanding the task instead of processing excessive information.
- Familiar visual theme increases engagement and motivation.

FRAME 2: INSTRUCTION PAGE



Analysis of the Instruction Screen

This screen explains how the memory game works and prepares the user for the observation and recall phases.

1. Purpose of the Screen

- Clearly communicates game rules and mechanics.
- Ensures users understand the memory challenge before starting.

2. Chunking of Instructions

- Instructions are presented in **numbered bullet points**.
- Each instruction focuses on a single action:
    - Memorize the elements
    - Observe carefully within the time limit
    - Elements will be hidden
    - Match the correct tiles
- A **special rule (Guardian Icon)** is separated as its own chunk to highlight importance.

3. Visual Hierarchy

- “INSTRUCTIONS” heading is bold and centered.
- Spacing between bullet points improves readability.
- **CONTINUE** button is visually distinct and placed at the bottom.

4. Cognitive & UX Benefits

- Chunked instructions reduce memory strain.
- Step-by-step flow improves task clarity.
- Highlighting special rules improves recall during gameplay.

FRAME 3: GAME SCREEN – OBSERVATION PHASE

Analysis of the Observation Phase Screen

This screen represents the **encoding phase**, where users visually observe and memorize elements.

1. Purpose of the Screen

- Allows users to view all visual elements before they are hidden.
- Initiates short-term memory encoding.

2. Chunking in Visual Design

- Icons are arranged in a **grid layout**, forming natural visual chunks.
- Similar icons (hats, crystals, food items) appear multiple times, encouraging pattern-based grouping.
- Time limit (“30 Secs”) is isolated at the top, separate from the grid.

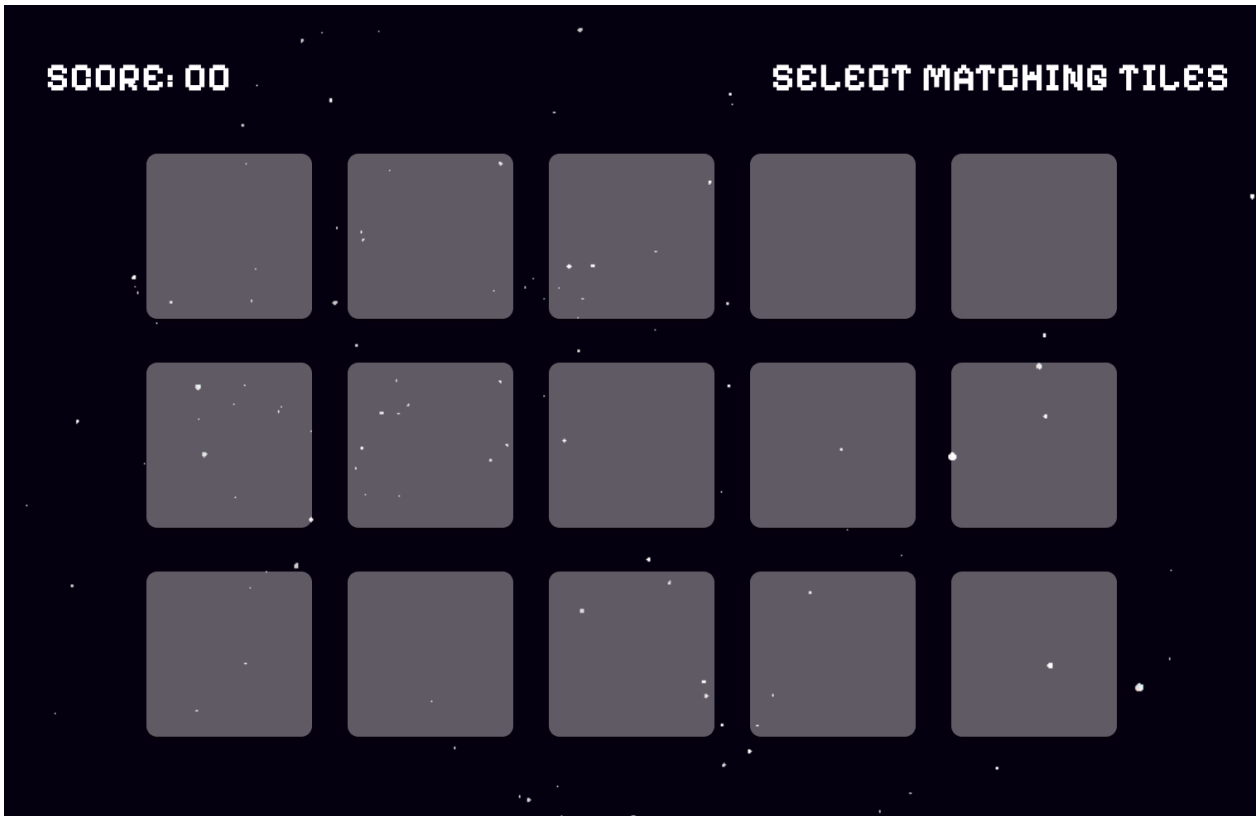
3. Visual Design Elements

- Equal spacing between tiles ensures visual balance.
- Icons are consistent in size and style.
- Countdown timer creates urgency without clutter.

4. Cognitive Benefit

- Grouping icons into a grid supports chunk-based memory encoding.
- Users remember icon **positions + patterns**, not isolated elements.
- Enhances spatial and visual memory.

FRAME 4: GAME SCREEN – RECALL PHASE



Analysis of the Recall Screen

This screen evaluates the user's ability to retrieve memorized information.

1. Purpose of the Screen

- Tests visual memory recall after hiding elements.
- Measures accuracy of remembered icon positions.

2. Chunking Strategy

- Grid layout remains consistent with the observation phase.
- Blank tiles act as placeholders, reinforcing spatial memory.
- Revealed matches appear in pairs, strengthening chunk-based recall.

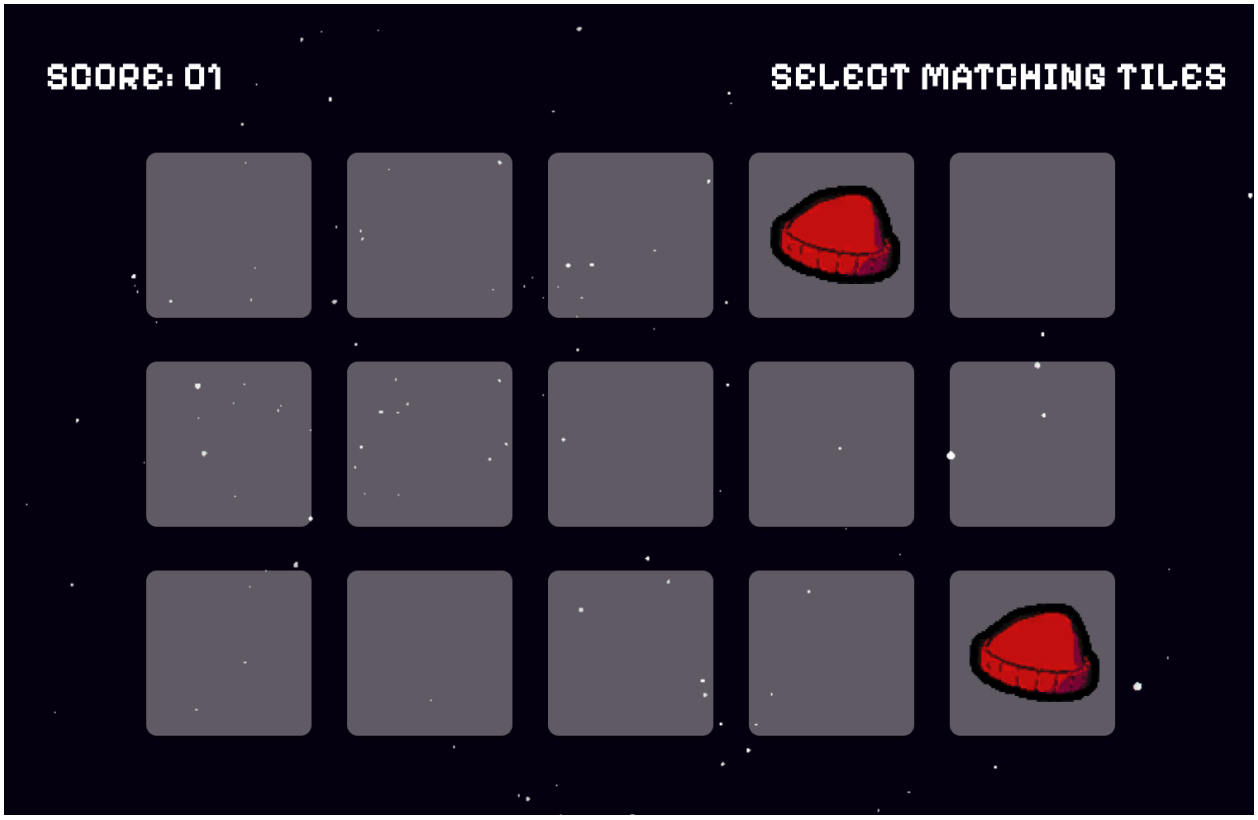
3. Interaction Design

- Users click tiles to reveal icons.
- Score counter updates dynamically.
- Guardian icon rule allows clearing multiple tiles, reinforcing learned patterns.

4. Cognitive & UX Benefits

- Consistent layout reduces re-learning effort.
- Encourages recall based on grouped memory rather than trial-and-error.
- Immediate visual feedback supports learning.

FRAME 5: FINAL SCORE / FEEDBACK SCREEN



Analysis of the Score Feedback Mechanism

This stage provides performance evaluation to the user.

1. Purpose of the Screen

- Displays user's memory performance.
- Concludes the experiment with measurable results.

2. Score Presentation

- Score shown numerically (e.g., Score: 01).
- Clear text ensures instant understanding.
- Minimal elements keep focus on results.

3. Feedback & Motivation

- Simple presentation avoids unnecessary distraction.
- Encourages replay to improve memory performance.
- Reinforces learning through outcome visibility.

4. Cognitive & UX Benefits

- Immediate feedback strengthens memory reinforcement.
- Helps users reflect on recall effectiveness.
- Supports learning through repetition.

Conclusion

The designed UI effectively demonstrates the role of **chunking** in improving visual memory recall. By grouping visual elements, maintaining consistent layouts, and presenting information in structured segments, the interface reduces cognitive load and enhances user performance. The game-based approach increases engagement while serving as an effective tool for evaluating memory recall.

Prototype Link

https://www.figma.com/design/WaWVccTQVJpF4dZY0rSX0U/exp1?node-id=0-1&t=FUV8vh35NtT14YtX-1